

| Question number | Answer                                                                                                                                                                                                                                                                                                          | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 4(a)            | <p>An answer that includes two of the following points:</p> <ul style="list-style-type: none"> <li>• higher the mean resting heart rate the lower the mean life span (1)</li> <li>• correct conclusion for mean resting heart rate data (1)</li> <li>• correct conclusion for mean lifespan data (1)</li> </ul> | <p>accept converse<br/>accept negative correlation between resting heart rate and mean life span</p> <p>e.g. the mouse has the highest (mean resting) heart rate / the whale has the lowest (mean resting) heart rate</p> <p>e.g. humans have the longest life span / mouse has shortest lifespan / human life span is anomalous</p> <p><b>Do not</b> accept conclusions based on size of animals (as no data)</p> <p>Accept as mean resting heart rate decreases life span increases except for humans = 2 marks</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                             | Additional guidance | Mark |
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| 4(b)(i)         | <p>Choose an item.</p> <p>B is the correct answer</p> <ul style="list-style-type: none"> <li>A is not the correct answer as 6bpm is not the heart rate shown in the trace</li> <li>C is not the correct answer as 60bpm is not the heart rate shown in the trace</li> <li>D is not the correct answer as 75bpm is not the heart rate shown in the trace</li> </ul> |                     | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                    | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                   | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 4(b)(ii)        | <p>An answer that includes two of the following points:</p> <ul style="list-style-type: none"> <li>{Smaller / shorter} P wave in (trace) A (1)</li> <li>R peak sharper in (trace) A (1)</li> <li>shorter QRS wave duration in (trace) A (1)</li> <li>T wave {larger / longer} in (trace) A (1)</li> </ul> | <p>accept converse unless refer to labels PQRST they will not get the marks</p> <p>accept one R peak in (trace) A and two R peaks in (trace) B<br/>accept QRS for R peak</p> <p>accept B (trace) has a larger hyperpolarisation / S is {larger / goes further down / wider} in B</p> <p>ignore T goes down in (trace) B<br/>accept the {trace / S-T} goes below the line in B (whereas A doesn't)</p> | (2)  |

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| 4(c)(i)         | <p>An answer that includes two of the following points:</p> <ul style="list-style-type: none"> <li>• (improved) {health care / vaccines / antibiotics / medication / diet} (1)</li> <li>• skills and knowledge in disease control (1)</li> <li>• failure in cardiovascular system may not be fatal as it is for most mammals (1)</li> </ul> | <p>ignore ref to {myogenic / live in better habitat / intelligence}</p> <p>accept medication to control defects<br/>accept can adjust lifestyle to reduce risk of disease<br/>what the person can do for themselves</p> <p>accept technology e.g. surgery / stent for CVD<br/>accept transplants<br/>what others need to do to them</p> <p>accept ref to any named organ / system<br/>accept no death due to predators<br/>ignore no predators<br/>humans don't have to run to catch their own food (or another reason linked to HR staying lower {so don't use up the number of beats too quickly / less damage to cardiac muscle})</p> | (2)  |

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| 4(c)(ii)        | <p>An answer that includes one of the following points:</p> <ul style="list-style-type: none"> <li>cardiac output = heart rate x stroke volume (1)</li> <li>cardiac output gives an indication of the efficiency of the heart / it is a measure of the rate at which blood is pumped {through heart / around body} (1)</li> </ul> | <p>accept cardiac output also includes stroke volume<br/>accept different animals may have different sized {hearts / ventricles}</p> <p>accept volume of blood pumped {per minute / per unit time}<br/>accept volume of blood per beat</p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Additional guidance                                                                                                                                                                                                                                                | Mark |
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| 4(c)(iii)       | <p>A description that includes four of the following points:</p> <ul style="list-style-type: none"> <li>cardiac output increases (1)</li> <li>(because increased respiration leads to) {increased carbon dioxide / decrease in pH} (in the blood) which is detected by chemoreceptors (1)</li> <li>(which) send impulse(s) to {cardiac control centre / medulla (oblongata)} (1)</li> <li>(increased) impulses through {sympathetic nervous system / neurones} (1)</li> <li>(increased) impulses to SAN which {depolarises more frequently / increases heart rate} (1)</li> <li>(impulses to SAN which) increases force of contraction <b>of cardiac</b> muscle / causes <b>cardiac</b> muscles to contract harder (1)</li> </ul> | <p>ignore signals</p> <p>accept adrenaline released from adrenal gland<br/>accept baroreceptors detecting increase in blood pressure</p> <p>accept (adrenaline) travels in the blood</p> <p>accept (adrenaline) binds to SAN which depolarises more frequently</p> | (4)  |

